SMART INDIA HACKATHON 2026

Problem Statement ID - 26231

Problem Statement Title - Digital Companion for Field Drug Testing

Theme - MedTech / BioTech / HealthTech

PS Category - Software

Team ID - K26234

Team Name - HexTech

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Proposed Solution

FieldProof: any phone + a ₹10 colour card = an objective drug-test result, sealed as tamper-proof evidence.

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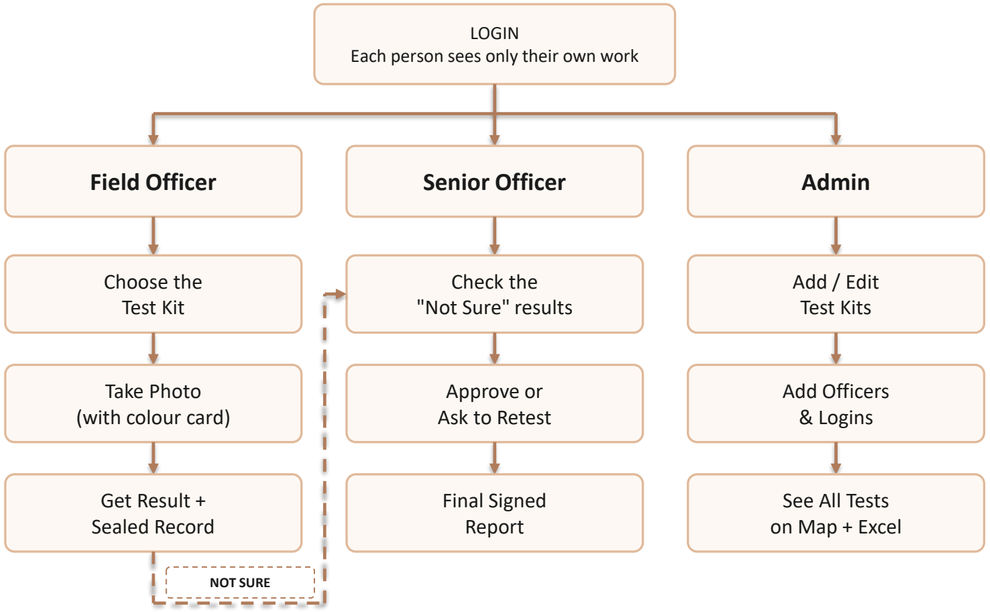
Problem:

- Today, police check the drug-test kit colour with their own eyes. Different officer or different light = different answer.
- There is no proof of when, where and by whom the test was done. So the result cannot be used as evidence in court.

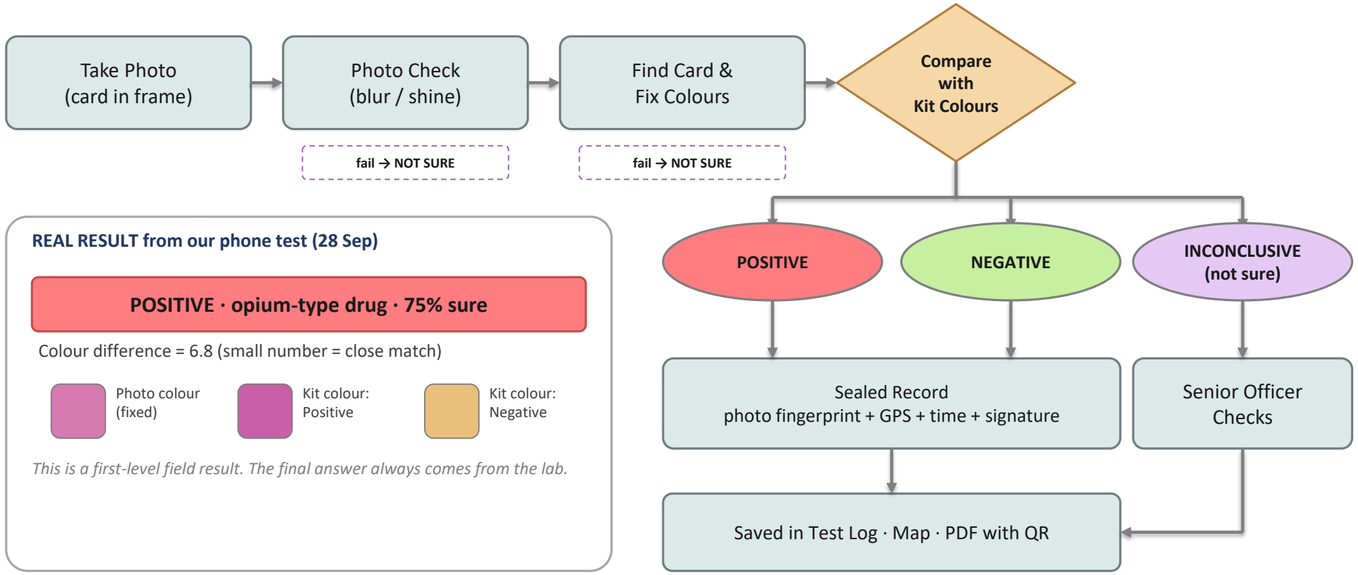
Innovation & Uniqueness (what is new):

- **App reads the colour, not the eye -** a ₹10 printed colour card fixes the light, and the app gives a colour score.
- **Honestly says "Not Sure" -** blurry photo, shine or unclear colour → INCONCLUSIVE with the reason; a senior officer checks.
- **Proof that cannot be faked -** photo fingerprint + GPS + internet time + officer ID, digitally signed; check by scanning QR.
- **Works with any test kit -** new kits are added from the website with their colours. No app update needed.

WHO DOES WHAT (ROLE-BASED WORKFLOW)



HOW THE APP DECIDES THE RESULT



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TECHNICAL APPROACH

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Technology Workflow (step by step)

1. Take Photo
Flutter mobile app
(guide + timer)

→ **2. Photo Check**
OpenCV checks
blur / shine

→ **3. Find the Card**
4 corner markers
(ArUco)

→ **4. Fix the Colours**
using the card's
known colours

↓

5. Decide Result
colour difference
score (ΔE)

← **6. Seal the Record**
fingerprint + GPS
+ time + signature

← **7. Save on Server**
FastAPI +
PostgreSQL

← **8. Officer Website**
React + map
+ PDF / Excel

How the App Decides (never guesses)

1. All 4 card corners found? → No → NOT SURE
2. Photo sharp, no shine? → No → NOT SURE
3. Closest kit colour: difference ≤ 14? → No → NOT SURE
4. Clear winner: 2nd best ≥ 4 further? → No → NOT SURE

Yes → POSITIVE / NEGATIVE + "% sure" (limits are set per kit)

How the ₹10 Colour Card Works

Printed Colour Card

- **4 corner markers -** app finds and straightens the card
- **24 boxes of known colours -** app learns how the light changed colours
- **Test window -** app fixes the kit colour and gives a score

Add Any Test Kit

Admin adds a kit with its result colours from the website. No app update needed.

Admin Adds Kit → Enter Result Colours → Set Match Limit → Ready in App

Any phone, any light → same fixed colour | Card not read properly → NOT SURE

Proof That Cannot Be Faked

Take Photo → Original Photo

↓

GPS + Internet Time + Officer + Phone ID → Photo Fingerprint (SHA-256) + Signature

↓

Chained Test Log → QR Check Page

Change even 1 byte → QR page shows TAMPERED

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FEASIBILITY AND VIABILITY

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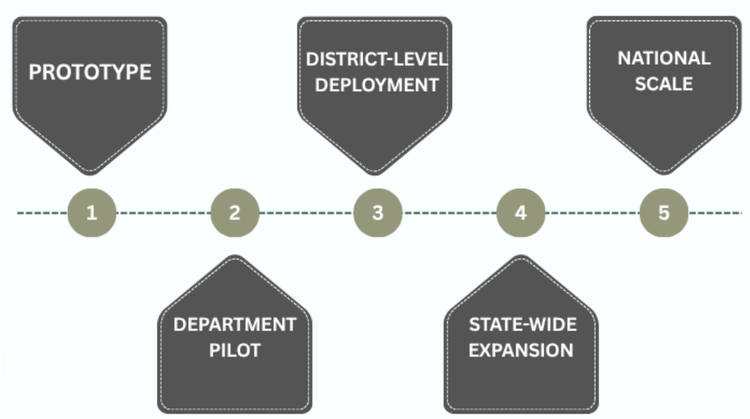
Feasibility (can we really do it?)

- **Technical -** built on our old project's tested code + OpenCV for colours.
- **Easy to use -** one photo, result in seconds; card can be printed on the kit pouch.
- **Low cost -** phone + ₹10 printed card + free software. No new machine.
- **Data plan -** each kit calibrated with ~250 real photos (5 lights × 50) at an FSL.

Potential Challenges & Strategies

Real-world problem	What our app does
Bad light (street lamp / torch)	Colour card fixes the colours
Shiny or blurry photo, card missing	Says NOT SURE and asks to retake
Colour is in between two results	Says NOT SURE; senior officer checks
Fake photo or edited record	In-app camera only + photo fingerprint
Kit from another maker or batch	Own profile per maker/batch; else NOT SURE

Phased Implementation & Scalability



Our Goals for the Prototype

- **Accuracy:** 90%+ correct on our demo colour samples in 4 types of light
- **Speed:** result + sealed record in 10 seconds or less
- **Tamper check:** change even 1 byte → check fails
- **Honesty:** never guesses - unclear means NOT SURE

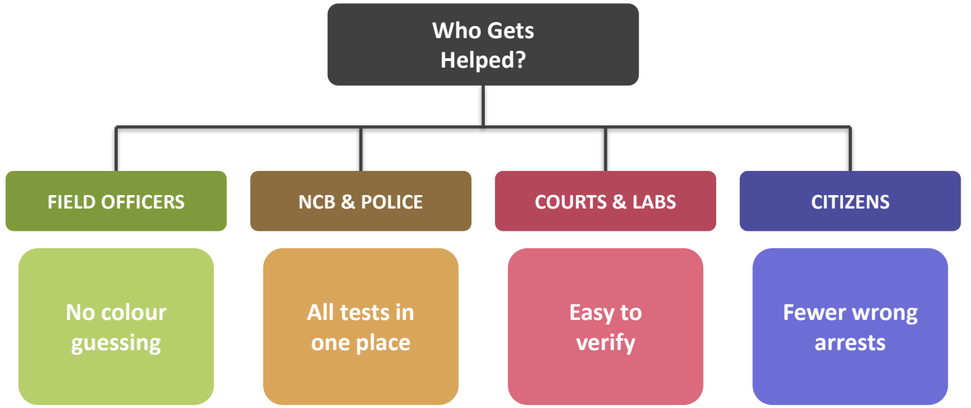
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IMPACT AND BENEFITS

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Benefits

- Same result, whoever does the test
- Proof for field tests in ~1.2 lakh NDPS cases a year (NCRB 2023)
- Fewer wrong cases - app says "not sure" instead of guessing
- No paperwork: PDF with QR, Excel, search and map
- Follows new laws: BNSS Sec 105 and BSA Sec 63



Expected Impact

- **Field Officer -** quick and confident, no colour guessing
- **NCB / Police -** every test saved, searchable, on a map
- **Courts / Labs -** scan QR to check the record is real
- **Common people -** less chance of wrong arrest

Future Scope (next steps)

- **Works offline -** at highways / borders, sends data later
- **Phone security chip -** phone also signs every test
- **Link with police systems -** e-Sakshya and CCTNS
- **Match with lab results -** find which kits are most accurate
- **Hindi + regional languages -** with voice help
- **Card on the kit pouch -** no loose card to carry

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RESEARCH AND REFERENCES

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#	Research (click the title to open)	What they found	How we use it
1	**IEEE EMBC 2016** Illumination and device independence for colorimetric detection with smartphone (Karisen & Dong)	Phone camera colours change with light and phone model; a reference fixes this	**Our ₹10 colour card fixes colours in every photo**
2	**IEEE TPS-ISA 2024** Digital Evidence Chain of Custody: Navigating New Realities of Digital Forensics (Nath et al.)	Digital evidence must prove it was never changed, and who handled it	**Photo fingerprint + signature + chained log**
3	**Forensic Science International 2014** Real-time quantitative colourimetric test for methamphetamine using mobile phone technology (Choodum et al.)	A phone camera can measure drug-test colours as numbers	**Our app reads the test colour as a number**
4	**Analytical Chemistry 2016** Objective presumptive field-testing of illicit drugs using smartphone analysis (Krauss et al.)	Reading test colours by eye is a personal guess	**App gives POSITIVE / NEGATIVE / NOT SURE**
5	**Digital Discovery 2023** Teaching old presumptive tests new digital tricks with computer vision	Different people describe the same colour differently	**App shows photo colour next to kit colours**
6	**Analytical & Bioanalytical Chemistry 2025** RGB colour correction and gamut limitations in smartphone-based analysis (Fyfe et al.)	Light and camera angle change colours; colour correction fixes most of it	**Colour fixing + NOT SURE if fixing fails**
7	**Forensic Science International 2005** False positives and false negatives with a cocaine-specific field test (Tsumura et al.)	Field colour tests can give wrong results	**App says NOT SURE; final answer from the lab**

Official sources: NDPS Act & Sampling Rules 2022 · NCB Field Officers' Handbook · UNODC Drug Field-Testing Manual · BNSS / BSA 2023 (MHA)

Competitor Map - where we stand

Option	Works with existing kits	No new hardware	Objective reading	Signed proof
Officer's eye (today)	✓	✓	✗	✗
DetectaChem app (USA)	✗	✓	✓	~
TruNarc / Raman device	✓	✗	✓	~
e-Sakshya app (MHA)	✓	✓	✗	✓
FSL lab test (GC-MS)	-	✗	✓	✓
FieldProof (ours)	✓	✓	✓	✓

✓ yes ~ partly ✗ no

Our edge: not just a colour reader.
The signed, chained evidence record
is the product - checkable by anyone.

Prototype Demo Video: Prototype Video Link

Take photo → Fix colour → Get result → Keep proof